ADVANCED PROGRAMMING ASSIGNMENT

Name: Gourisankar Bhuyan
Roll No: CSB24065

Program Description:

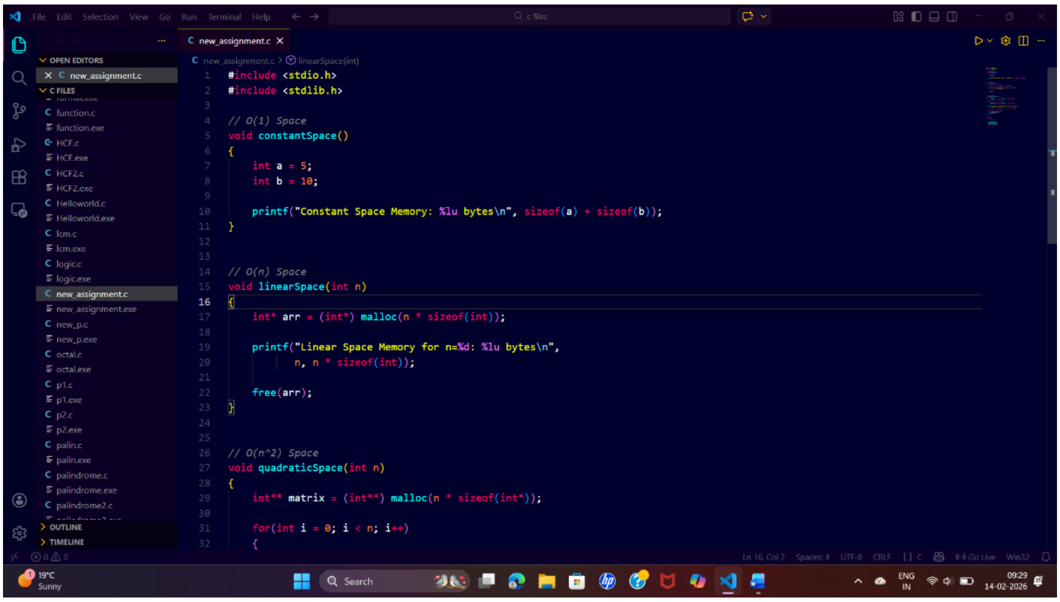
This program demonstrates different space complexities: Constant O(1), Linear O(n), and Quadratic O(n^2). It measures actual memory usage using sizeof() and dynamic memory allocation.

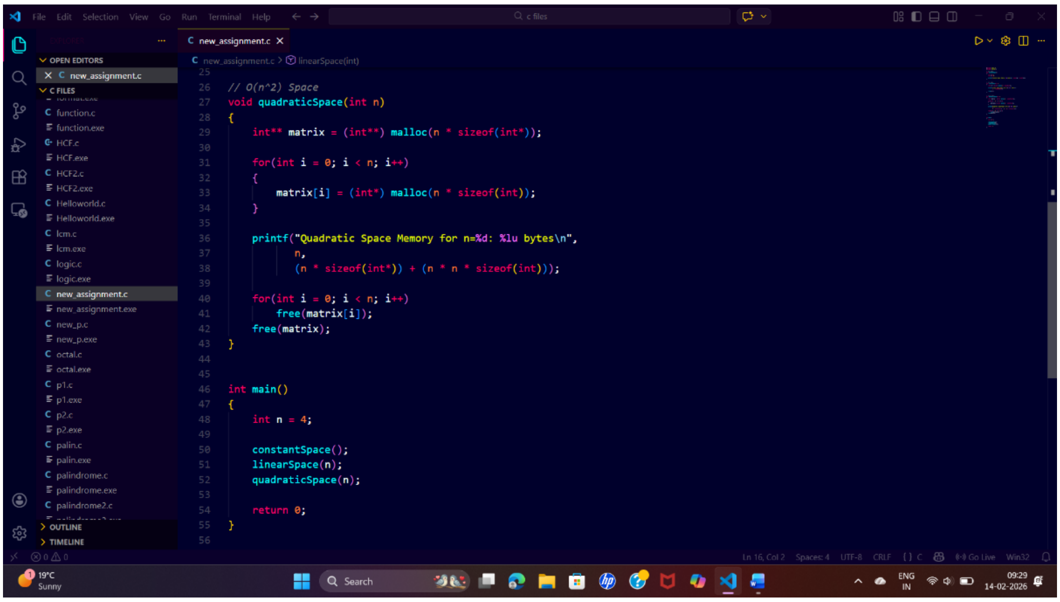
Source Code:

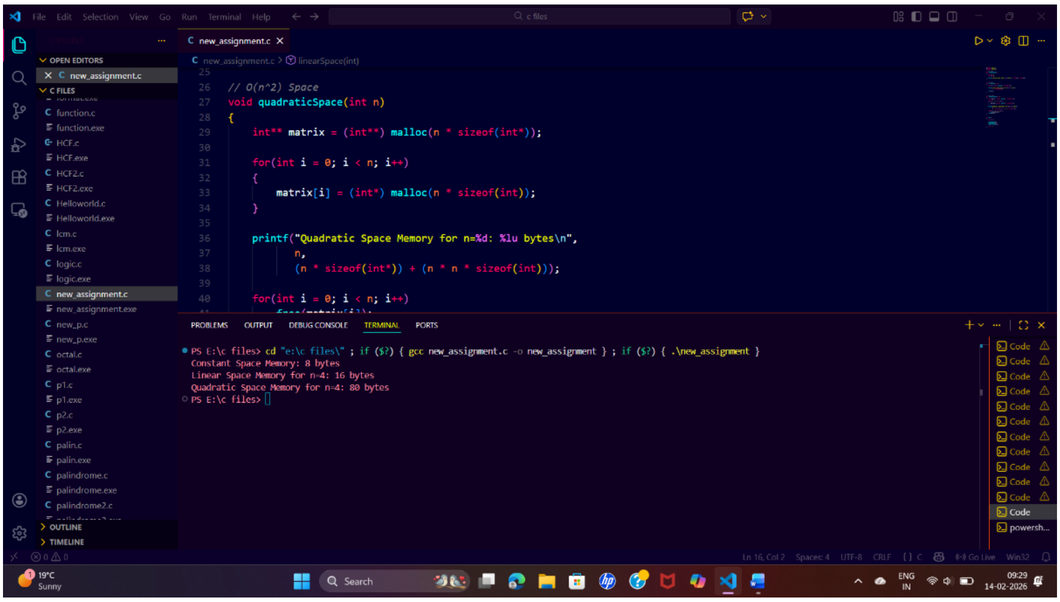
```
#include <stdio.h>
#include <stdlib.h>

// Function to demonstrate O(1) Space Complexity
void constantSpace() {     int a = 5;     int b
= 10;

    printf("Constant Space Memory: %lu bytes\n", sizeof(a) + sizeof(b));
}

// Function to demonstrate O(n) Space Complexity
void linearSpace(int n) {     int *arr;

    arr = (int *)malloc(n * sizeof(int));

    printf("Linear Space Memory for n = %d: %lu bytes\n",
n, n * sizeof(int));

    free(arr);
}

// Function to demonstrate O(n^2) Space Complexity
void quadraticSpace(int n) {     int **matrix;

    matrix = (int **)malloc(n * sizeof(int *));

    for (int i = 0; i < n; i++)     {
matrix[i] = (int *)malloc(n * sizeof(int));     }

    printf("Quadratic Space Memory for n = %d: %lu bytes\n",
n,            (n * sizeof(int *)) + (n * n * sizeof(int)));

    for (int i = 0; i < n; i++)
{         free(matrix[i]);
}

    free(matrix);
}
```

```
int main() {
int n = 4;

constantSpace();
linearSpace(n);
quadraticSpace(n);

    return 0;
}
```

Screenshots of the actual program & it’s outputs:




new_assignment.c
// O(n^2) Space
void quadraticSpace(int n)
{
int** matrix = (int**) malloc(n * sizeof(int*));
for(int i = 0; i < n; i++)
{
matrix[i] = (int*) malloc(n * sizeof(int));
}
printf("Quadratic Space Memory for n=%d: %lu bytes\n",
n,
(n * sizeof(int*)) + (n * n * sizeof(int)));
for(int i = 0; i < n; i++)
free(matrix[i]);
free(matrix);
}
int main()
{
int n = 4;
constantSpace();
linearSpace(n);
quadraticSpace(n);
return 0;
}


PS E:\c files> cd "e:\c files\" ; if ($?) { gcc new_assignment.c -o new_assignment } ; if ($?) { .\new_assignment }
Constant Space Memory: 8 bytes
Linear Space Memory for n=4: 16 bytes
Quadratic Space Memory for n=4: 80 bytes
PS E:\c files>